And the Race goes on...

The wait for GeForce3 was definitely worth it and the card lived up to everybody's expectations. For the past few months we would say the name GeForce3 is on everybody's lips and everyone in the 3D gaming world is treating it like nothing less than god. So the GeForce3 is rocking and all set to rule the 3D accelerator scene…. Well, ATi thinks otherwise and is all ready to give the GeForce3 the stick.

Welcome the ATi Radeon 8500, based on the much-awaited R200 chipset. Looking at the feature list of this card, it seems that we will be coming one step closer to movie-quality special effects. The card will incorporate a new technology called Truform from ATi. Other additions include Smartshader, Smoothvision, HyperzII, Hydravision, Charisma EngineII and Pixel TapestryII, just to name a few. The chip will be built on the 0.15-micron process and will be crammed with 60 million transistors.

Finally, the GeForce3 will be feeling the heat and the hard-core gamers will have a choice. The card will cost $399 (Indian launch prices may be much higher) and should be available by September this year.

However, nVidia is in no mood to hit the brakes. Here are the rumoured specifications for the, hold your breath… GeForce4. The core frequency will probably run between 250-300 MHz and the memory will also run somewhere in the same range (DDR of course). Texture swapping will be a cakewalk with 128 MB RAM, which will now be the default specification. The chip will feature two vertex shaders as in the Xbox—the GeForce3 only has one. With a fill rate of 2.4 gigatexels per second, which is almost double that of the GeForce3, and a memory bandwidth of 9.6 GBps, this chip will leave only one place for the chips of today…the history books.

In the more immediate future, going by the nVidia way of coming out with products every six months, we could expect a GeForce3 Ultra very soon. The specifications could be on the 0.15-micron process with 128 MB DDR RAM. The GeForce3 Ultra could offer a memory bandwidth of a cool 8 GBps, running at a memory frequency of 250 MHz DDR.

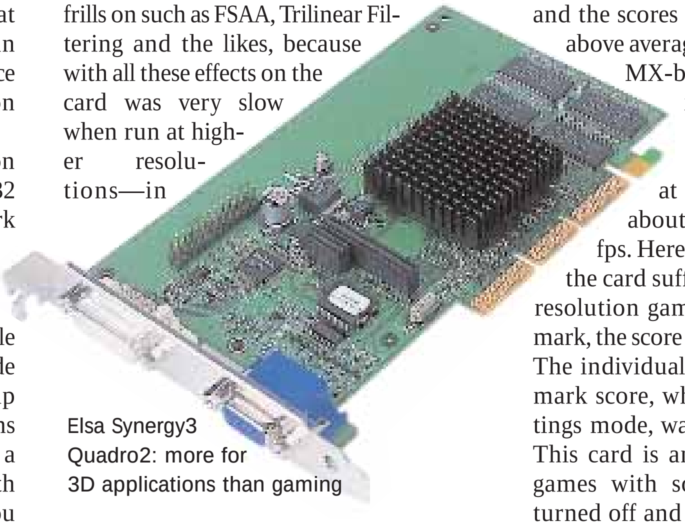
The competition in the 3D card scene is hotting up like never before and the two biggest chip manufacturers, nVidia and ATi, are all set to fight it out. Which spells good news for the gaming enthusiasts.

score of 5529. The scores in *Quake III* were also impressive: with the normal settings, it gave an average of 192.8 fps. The card features the Quadro2 MXr chipset, which is the nVidia version of a workstation solution. However, the card is highly recommended if you are less into gaming and if you work more with professional 3D applications. Through this card you can attach two monitors, be it analog, digital, VGA or DVI. This could be extremely useful if you work with 3D packages such as 3D Studio Max, Maya, Light Wave, etc. Its price at Rs 30,000 will definitely burn a hole in your pocket but if you don't mind the price and need a good card for your workstation needs, this card is worth a try.

The Matrox G450 32MB is based upon the G450 processor. This card features 32 MB of onboard memory. In the 3DMark 2000 test, it logged a score of 2380, which was just a shade higher than its 16 MB counterpart. In *Quake III Arena*, with normal settings we logged a very playable 73 fps with this card. But in the max mode (1280x1024 at 32-bit depth) it could pump out only an average of 10 fps, which means that if you don't mind playing games at a lower resolution of 800x600 at 16-bit depth with some of the effects turned off then you could consider this card.

The scores in the *Serious Sam* test were also very low: with low-quality visual settings, the average frame rate was a mere 26.5 and with the max settings, we logged just 7.3 fps. This again proves that this card is not suitable for playing games at high resolution with 32-bit textures on.

The Asus GeForce2 MX200 Magic features 32 MB of video memory with a 350 MHz RAMDAC, allowing for very decent gameplay. In the 3DMark 2000 test, it gave a score of 3243 on our high-end machine, which is a good score for cards in its class. The performance of this card makes it suitable for the average gamer. The *Quake III* tests show that the card is not suitable for playing high-end games with all the frills on such as FSAA, Trilinear Filtering and the likes, because with all these effects on the card was very slow when run at higher resolutions—in the max settings test, it returned a score of just 13.7 fps. Even in the *Serious Sam* test, the card returned a playable score of 74.5 fps in the low settings mode, but died out in the max settings mode (average frame rate of 7.5 fps).



Elsa Synergy3 Quadro2: more for 3D applications than gaming

The AOpen PA256 GeForce2 MX features the GeForce2 MX processor, which is a scaled-down version of the GeForce2 GTS chip. It has a memory bandwidth of 2.8 GBps, which would easily get you gaming in low to medium resolutions with texture and colour depth settings set to 16 bit. If you plan to play at high resolutions with textures and colour settings at 32 bit, then you should avoid an MX-based solution.

The performance of this card was good and the scores in the *Quake III* tests were above average when compared to other MX-based solutions. At settings, it gave an average frame rate of 180 fps and with max settings (1280x1024 at 32-bit depth) it gave a just about playable frame rate of 29.5 fps. Here you can easily observe that the card suffers when it comes to high-resolution gaming. In the *Evolva* benchmark, the score was 52.5 fps, which is good. The individual Vulpine GL Mark benchmark score, when tested in the max settings mode, was quite low (a mere 13.9). This card is an ideal choice if you play games with some of the extra effects turned off and at a medium resolution of 800x600 at 16-bit depth.

Asus V3800 Riva TNT2 Pro, with 140/166 MHz core and memory, which is higher than 125/150 MHz core and memory offered by the TNT2, is one of the older cards that are still on the scene and is